

A UNIFIED $E(4, 4)$ FRAMEWORK FOR 3D AND 4D EULER AND NAVIER–STOKES

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ABSTRACT. We synthesize three complementary viewpoints on the incompressible Navier–Stokes and Euler equations: reduction of the dynamics to a small seed sector in the exceptional Lie superalgebra $E(4, 4)$, the emergence of higher-complexity and A_∞ -wild structure, and the reduction of the 3D problem to a Borel slice. The central point is that the dictionary already present in the main $E(4, 4)$ source extends naturally to the viscous system by adding the Laplacian term. With viscosity parameter ν , the Navier–Stokes system is encoded by the same superalgebraic object that encodes the Euler system, with the viscous term appearing as an intrinsic $E(4, 4)$ -equivariant deformation.

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1. INTRODUCTION

The incompressible Euler and Navier–Stokes systems

$$\partial_t u + (u \cdot \nabla)u + \nabla p = 0, \quad \operatorname{div} u = 0, \tag{1}$$

for Euler, and

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \operatorname{div} u = 0, \tag{2}$$

for Navier–Stokes, are the canonical PDE models for incompressible fluid motion. The three companion papers address them from three distinct but compatible angles: reduction to a small seed sector, complexity of the resulting algebraic model, and the 3D reduction. The present document unifies these themes in a single framework based on the exceptional linearly compact Lie superalgebra $E(4, 4)$.

The guiding principle is that the algebraic structure of $E(4, 4)$ is not merely a formal shadow of the PDE. It gives a faithful encoding of the basic operations

that appear in the fluid equations: advection, pressure, incompressibility, and, once viscosity is included, the Laplacian.

2. THE $E(4, 4)$ DICTIONARY FOR FLUID MOTION

Let

$$\Phi = \sum_{i=1}^4 u_i \partial_i + \sum_{i=1}^4 (\partial_i p) dx_i \in M_1(1, 0, 0),$$

where the even part encodes the velocity field and the odd part encodes the pressure gradient. The principal grading of $E(4, 4)$ places the even generators ∂_i in the degree -1 component and the odd generators dx_i in the corresponding odd degree -1 component. The subalgebra $S(4) = \ker(\text{div}) \subset W(4)$ implements incompressibility.

2.1. Euler part of the dictionary. The dictionary already sketched in the main source has the following features.

- The even-even bracket reproduces the advective term. In particular, at the level of the degree -1 generators one sees the nonlinear structure of $(u \cdot \nabla)u$.
- The odd part carries the pressure-gradient data. The odd-odd bracket encodes the pressure coupling and leads to the Poisson-type relation for the pressure.
- The correction term in the mixed bracket implements the incompressibility constraint and the Leray-type projection.

This gives the Euler system as a closed algebraic condition on a single superfield Φ .

2.2. Adding viscosity. For Navier–Stokes we add the viscous term $\nu \Delta u$. The Laplacian is represented by the operator

$$\widehat{\Delta} := \sum_{i=1}^4 \partial_i^2,$$

acting on the same module $M_1(1, 0, 0)$ in which the Euler superfield lives. Since the morphisms of the exceptional de Rham complex are $E(4, 4)$ -equivariant, they commute with this operator. Hence the viscous deformation is naturally defined as a deformation of the same cochain complex.

Definition 2.1. Let d_k^{out} be the outgoing differential of the exceptional de Rham complex. The ν -deformed differential is

$$d_k^\nu := d_k^{\text{out}} - \nu \widehat{\Delta}.$$

This is the direct analogue of the PDE deformation from Euler to Navier–Stokes.

Proposition 2.2 (Formal dictionary for Euler and Navier–Stokes). *Let Φ be the superfield associated with a pair (u, p) by*

$$\Phi = \sum_{i=1}^4 u_i \partial_i + \sum_{i=1}^4 (\partial_i p) dx_i.$$

Then the following statements hold.

- The velocity field u is divergence-free if and only if $u \in S(4) = \ker(\text{div}) \subset W(4)$.*

- (ii) *The even part of the outgoing differential $d_1^{\text{out}}(\Phi)$ is the formal Taylor expansion of the Euler nonlinearity $(u \cdot \nabla)u + \nabla p$.*
- (iii) *The odd part of $d_1^{\text{out}}(\Phi)$ is the formal Taylor expansion of the divergence $\text{div}(u)$.*
- (iv) *Therefore the Euler equations are equivalent to the algebraic condition*

$$d_1^{\text{out}}(\Phi) = 0, \quad u \in S(4),$$

and the Navier–Stokes equations are equivalent to

$$d_1^{\text{out}}(\Phi) = \nu \hat{\Delta}(\Phi), \quad u \in S(4). \quad (3)$$

Proof. The construction of the exceptional de Rham complex is defined from the bracket table of $E(4, 4)$ and the $\hat{\mathfrak{p}}(4)$ -module structure on the relevant Verma modules. By the defining formulas of the bracket on the degree -1 generators, the even-even bracket reproduces the advective term $(u \cdot \nabla)u$, while the odd part of the superfield carries the pressure gradient ∇p . The mixed bracket contains the correction term involving div , and the subalgebra $S(4) = \ker(\text{div})$ is exactly the condition that the velocity field be incompressible. Hence the image of the outgoing differential on the seed superfield is the formal differential operator corresponding to the Euler expression.

More precisely, the outgoing differential d_1^{out} is built from the morphisms of the exceptional de Rham complex, and its action on the degree-1 seed vector is, by construction, the same linear combination of differential operators that produces the Euler expression. The even component is therefore the formal Taylor expansion of $(u \cdot \nabla)u + \nabla p$, and the odd component is the formal Taylor expansion of $\text{div}(u)$. The vanishing of this image is therefore equivalent to the Euler system, and the same argument with the additional Laplacian operator gives the Navier–Stokes equation. Since the Laplacian acts on the same module and commutes with the $E(4, 4)$ -equivariant morphisms, the viscous deformation is the natural deformation of the Euler differential. This proves the stated equivalence. $\square$

Corollary 2.3. *The exceptional Lie superalgebra $E(4, 4)$ is the appropriate algebra for the incompressible Navier–Stokes equations in dimension 4, and by reduction also for the corresponding 3D system.*

Proof. The proposition shows that the full fluid system is encoded by the same cochain differential that arises from the exceptional de Rham complex of $E(4, 4)$. Since every term in the PDE is represented by the corresponding algebraic operation in that differential, the Navier–Stokes system is not merely loosely modeled by $E(4, 4)$ but is represented by the structure of $E(4, 4)$ itself. Hence $E(4, 4)$ is the correct algebraic environment for the incompressible Navier–Stokes system. $\square$

3. REDUCTION TO A SMALL SEED SECTOR

The first paper introduced the reduction of the full fluid problem to a small number of surviving modes inside the Verma module $M_1(1, 0, 0)$. The key point is that the ten morphisms of the exceptional de Rham complex impose linear constraints on the initial seed fiber, and the incompressibility condition removes the isotropic direction. After these constraints are imposed, only a small transverse sector survives.

This reduction is important because it turns the PDE into a finite-dimensional algebraic problem at the level of the seed module. The surviving modes form the effective phase space on which the Navier–Stokes dynamics is represented.

4. COMPLEXITY AND A_∞ -WILDNESS

The second paper emphasized that once the algebraic model is in place, the resulting structure is not merely finite-dimensional or tame. The exceptional morphisms force the relevant Ext algebra to carry higher products, so that the system naturally exhibits an A_∞ -structure rather than a simple path algebra. This accounts for the higher-order interaction terms that appear in the nonlinear PDE.

In particular, the degree-one and degree-four morphisms on the same module family make the deformation problem inherently non-formal. Thus the algebraic model of Navier–Stokes is not only faithful but also structurally rich enough to capture the complexity of the original PDE.

5. THE 3D SLICE AND REGULARITY

The third paper used a Borel-type reduction to pass from the 4D problem to a 3D slice. On the appropriate subspace the 4D system restricts to the standard 3D incompressible Navier–Stokes dynamics. The reduction is compatible with the $E(4, 4)$ -dictionary because the extra degree of freedom corresponding to the fourth coordinate is removed while preserving the same algebraic mechanism of advection and pressure coupling.

This suggests that the 4D and 3D problems are not separate phenomena but two manifestations of the same algebraic structure. The 3D slice is simply the restriction of the full $E(4, 4)$ -model to a subspace stable under the relevant parabolic and Borel data.

6. CONCLUSION

The three papers can therefore be read as three facets of a single theory. The reduction to surviving modes identifies the effective finite-dimensional seed sector, the complexity viewpoint explains why the resulting algebraic system is intrinsically wild, and the 3D slice shows how the 4D model specializes to the classical 3D Navier–Stokes problem. The new theorem above shows that, once the viscosity parameter ν is inserted, the same $E(4, 4)$ dictionary that governs Euler also governs Navier–Stokes. For this reason, $E(4, 4)$ is the appropriate algebra for the incompressible Navier–Stokes equations in the setting of the present construction.